



Deep Learning

Lecture 6.2: Transformers in GPT and Vision

Today's Learning Objectives

Be able to answer:

- What are pretrained language representation (PRL) models?
 - How are language-model based PRL models trained?
 - What is GPT?
- How to use transformers to vision tasks?
 - What is ViT?
 - What is Swin Transformer?
 - What is DETR?
 - What is MaskFormer and Segment Anything (SAM)?





A sentence is a sequence of words $\mathbf{w} = w_1, w_2, w_3, \dots, w_T$ with $\forall w_i \in V$

Language model

A model that estimates the probability of natural language text.

$$P(\mathbf{w}) = P(w_1, w_2, w_3, \dots, w_T)$$

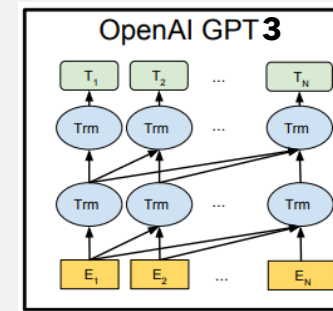
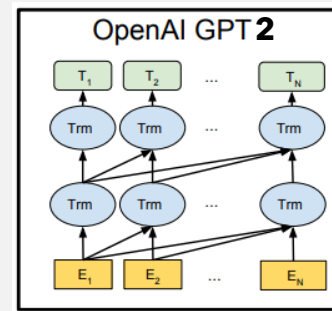
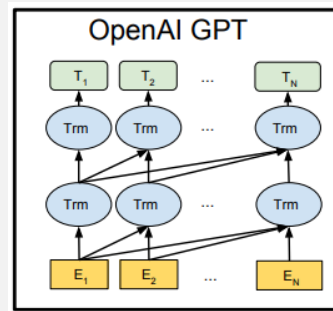
Often written

$$P(\mathbf{w}) = \prod_t \underbrace{P(w_t | w_1, \dots, w_{t-1})}$$

Probability of next word given history ... sound like predictive text?

What about just predict the next word at scale?

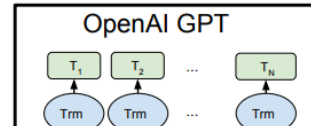
Unidirectional Language Models:





Generative Pre-trained Transformer (GPT)

Idea: Train a Transformer-based language model on a lot of data.



Improving Language Understanding by Generative Pre-Training

Alec Radford¹ Karthik Narasimhan¹ Tim Salimans¹ Ilya Sutskever¹
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Abstract

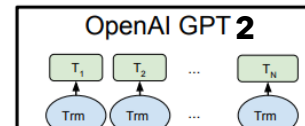
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1 Introduction

The ability to learn effectively from raw text is crucial to alleviating the dependence on supervised learning in natural language processing (NLP). Most deep learning methods require substantial amounts of manually labeled data, which restricts their applicability in many domains that suffer from a dearth of annotated resources [61]. In these situations, models that can leverage linguistic information from unlabeled data provide a valuable alternative to gathering more annotation, which can be time-consuming and expensive. Further, even in cases where considerable supervision is available, learning good representations in an unsupervised fashion can provide a significant performance boost. The most compelling evidence for this so far has been the extensive use of pre-trained word embeddings [10, 39, 42] to improve performance on a range of NLP tasks [8, 11, 26, 45]. Leveraging more than word-level information from unlabeled text, however, is challenging for two main reasons. First, it is unclear what type of optimization objectives are most effective at learning text representations that are useful for transfer. Recent research has looked at various objectives such as language modeling [44], machine translation [38], and discourse coherence [22], with each method outperforming the others on different tasks.¹ Second, there is no consensus on the most effective way to transfer these learned representations to the target task. Existing techniques involve a combination of making task-specific changes to the model architecture [43, 44], using intricate learning schemes [21] and adding auxiliary learning objectives [50]. These uncertainties have made it difficult to develop effective semi-supervised learning approaches for language processing.

¹<https://gluebenchmark.com/leaderboard>

Preprint. Work in progress.



Language Models are Unsupervised Multitask Learners

Alec Radford¹ Jeffrey Wu¹ Rewon Child¹ David Luan¹ Dario Amodei¹ Ilya Sutskever¹

Abstract

Natural language processing tasks, such as question answering, machine translation, reading comprehension, and summarization, are typically approached with supervised learning on task-specific datasets. We demonstrate that language models begin to learn these tasks without any explicit supervision when trained on a new dataset of millions of webpages called WebText. When conditioned on a document plus questions, the answers generated by the language model reach 55 F1 on the CoQA dataset - matching or exceeding the performance of 3 out of 4 baseline systems without using the 127,000+ training examples. The capacity of the language model is essential to the success of zero-shot task transfer and increasing it improves performance in a log-linear fashion across tasks. Our largest model, GPT-2, is a 1.5B parameter Transformer that achieves state of the art results on 7 out of 8 tested language modeling datasets in a zero-shot setting but still underfits WebText. Samples from the model reflect these improvements and contain coherent paragraphs of text. These findings suggest a promising path towards building language processing systems which learn to perform tasks from their naturally occurring demonstrations.

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Machine learning systems now excel (in expectation) at tasks they are trained for by using a combination of large datasets, high-capacity models, and supervised learning (Krizhevsky et al., 2012) (Sutskever et al., 2014) (Amodei et al., 2016). Yet these systems are brittle and sensitive to slight changes in the data distribution (Recht et al., 2018) and task specification (Kirkpatrick et al., 2017). Current systems are better characterized as narrow experts rather than

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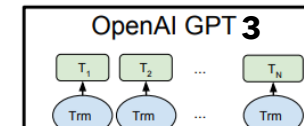
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The dominant approach to creating ML systems is to collect a dataset of training examples demonstrating correct behavior for a desired task, train a system to imitate these behaviors, and then test its performance on independent and identically distributed (IID) held-out examples. This has served well to make progress on narrow experts. But the often erratic behavior of captioning models (Lake et al., 2017), reading comprehension systems (Jia & Liang, 2017), and image classifiers (Alcorn et al., 2018) on the diversity and variety of possible inputs highlights some of the shortcomings of this approach.

Our suspicion is that the prevalence of single task training on single domain datasets is a major contributor to the lack of generalization observed in current systems. Progress towards robust systems with current architectures is likely to require training and measuring performance on a wide range of domains and tasks. Recently, several benchmarks have been proposed such as GLUE (Wang et al., 2018) and decaNLP (McCann et al., 2018) to begin studying this.

Multitask learning (Caruana, 1997) is a promising framework for improving general performance. However, multitask training in NLP is still nascent. Recent work reports modest performance improvements (Yogatama et al., 2019) and the two most ambitious efforts to date have trained on a total of 10 and 17 (dataset, objective) pairs respectively (McCann et al., 2018) (Bowman et al., 2018). From a meta-learning perspective, each (dataset, objective) pair is a single training example sampled from the distribution of datasets and objectives. Current ML systems need hundreds to thousands of examples to induce functions which generalize well. This suggests that multitask training may need just as many effective training pairs to realize its promise with current approaches. It will be very difficult to continue to scale the creation of datasets and the design of objectives to the degree that may be required to brute force our way there with current techniques. This motivates exploring additional setups for performing multitask learning.

The current best performing systems on language tasks



Language Models are Few-Shot Learners

Tom B. Brown^{*} Benjamin Mann^{*} Nick Ryder^{*} Melanie Subbiah^{*}
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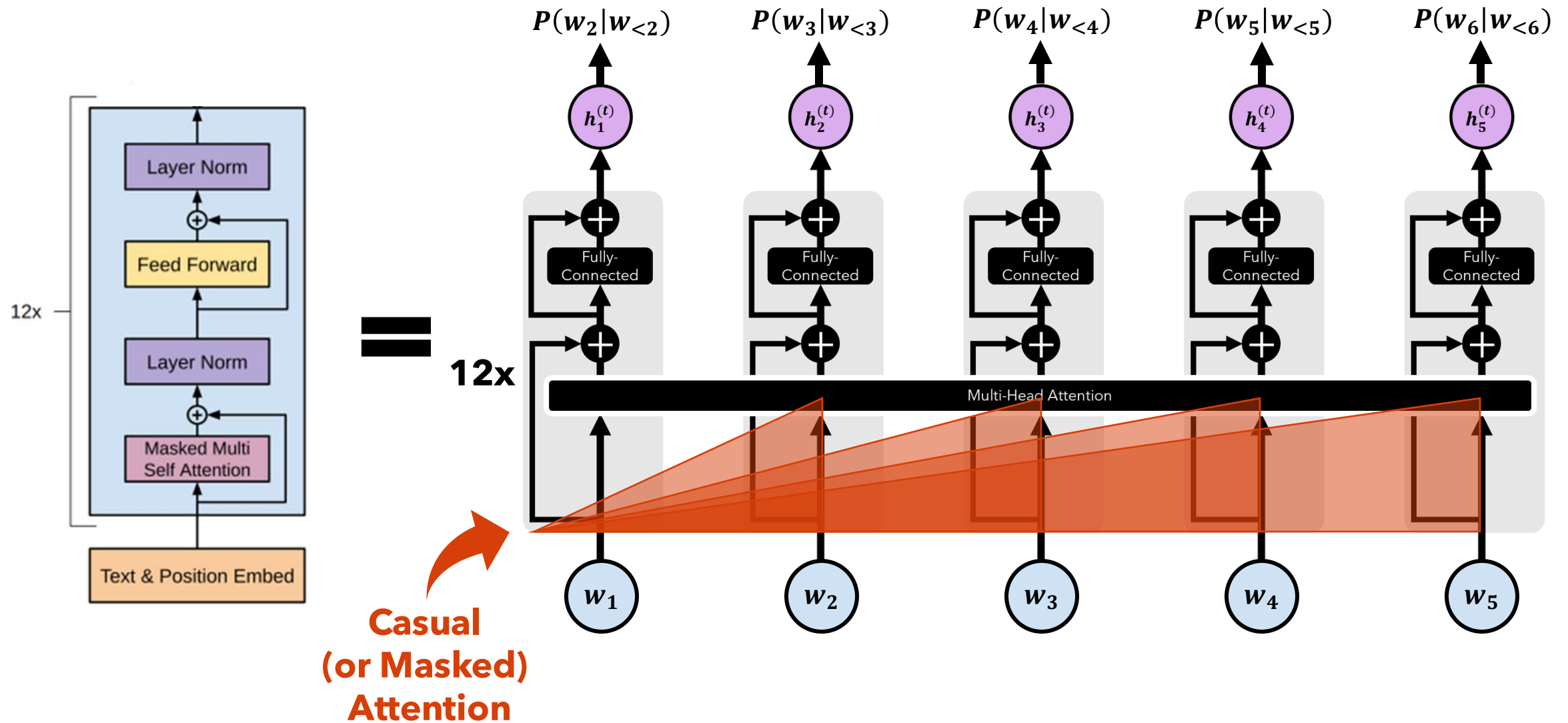
Author contributions listed at end of paper.

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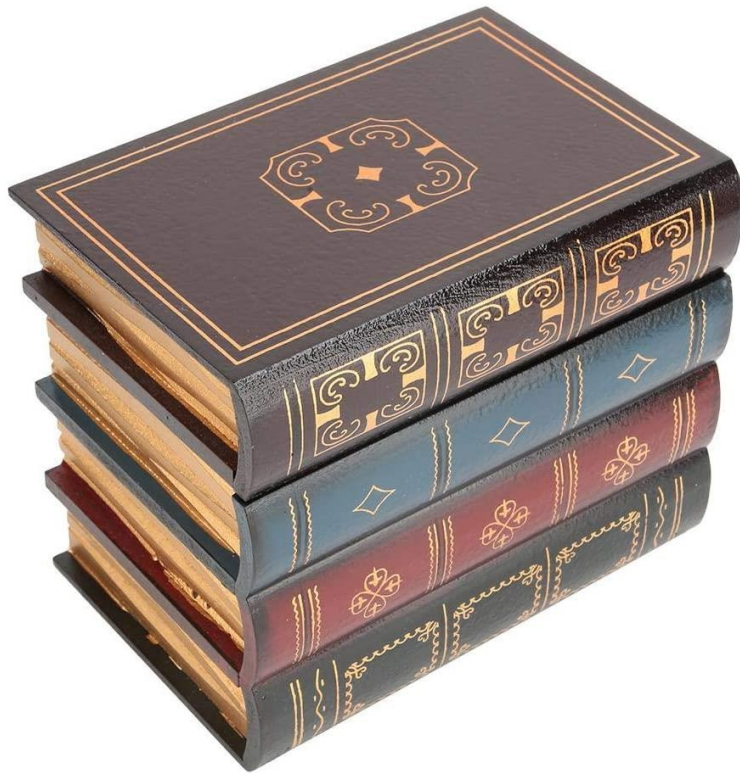
Generative Pre-trained Transformer (GPT)

Idea: Train a **Transformer-based language model** on a lot of data, then finetune.





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BookCorpus (Or Toronto Book Corpus)

- >7,000 unique books in various genres.
- And thus long stretches of contiguous text

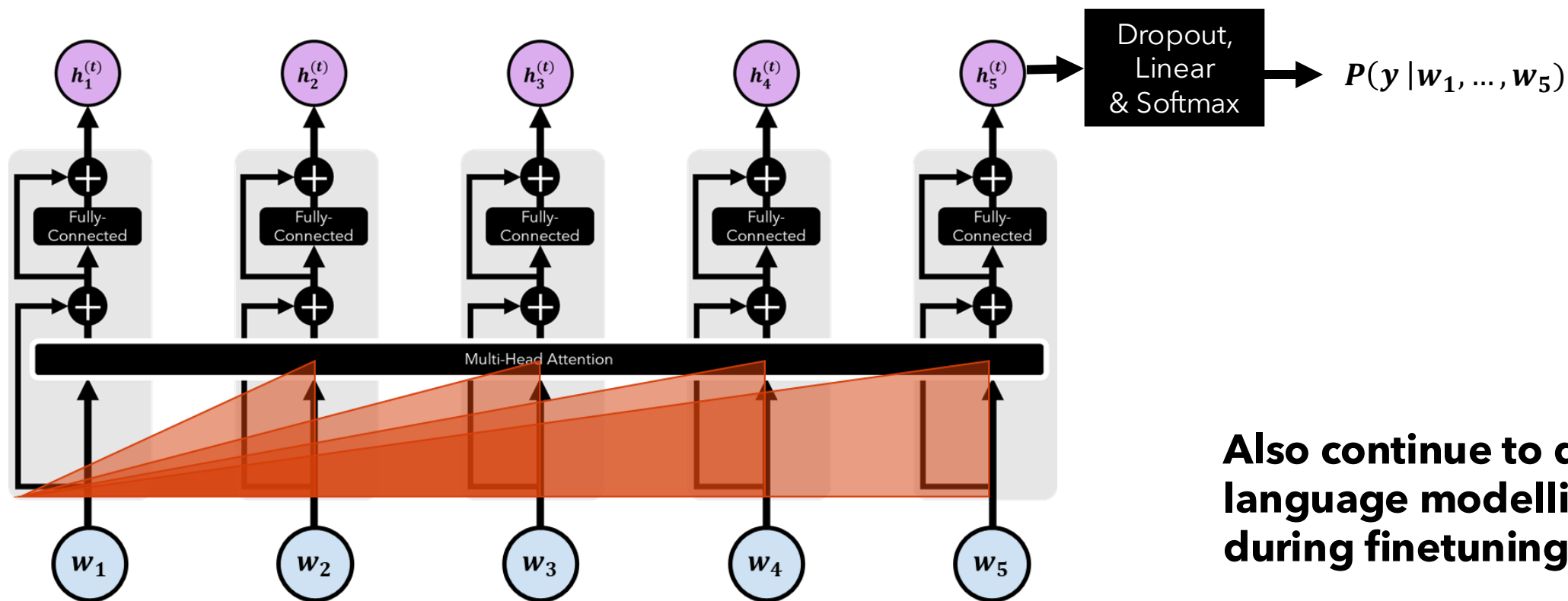
Aligning books and movies: Towards story-like visual explanations by watching movies and reading books.



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Pass representation of last token to a classifier:

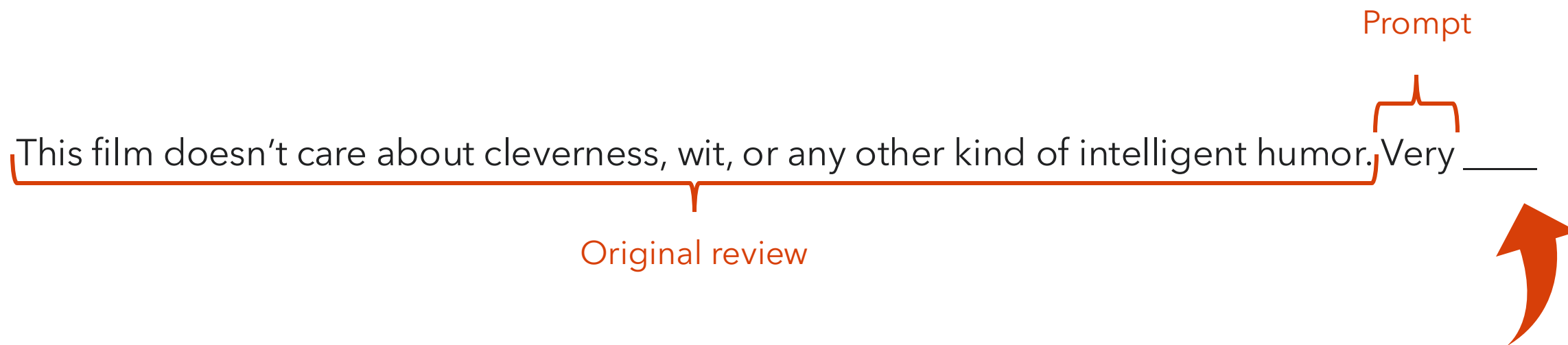


Also continue to do language modelling task during finetuning.



Interesting Bit: Zero-shot behaviors. A glimpse of things to come.

Design simple ways to get the language model to perform downstream tasks without training it for the task explicitly. Will use sentiment analysis as an example:

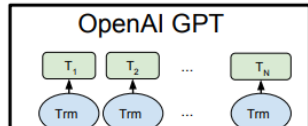


Predict instance as positive if the probability of word "positive" is higher than the word "negative".



Generative Pre-trained Transformer (GPT2)

Idea: Train a Transformer-based language model on a lot of data.



Improving Language Understanding by Generative Pre-Training

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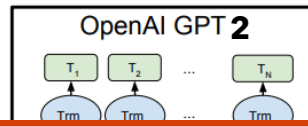
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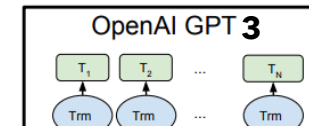
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Idea: Bigger model, more data, more zero-shot tests! **Everything is a LM task!**

Borrow notion from McCann et al, every NLP task can be expressed in language:

Sentiment
Analysis

A stirring, funny and finally transporting re-imagining of Beauty and the Beast and 1930s horror film. Is this sentence positive or negative? _____

Translation

Most of the planet is ocean water. What is the translation from English to German? _____

Question
Answering

Cycle of the Werewolf is a short novel by Stephen King, featuring illustrations by comic book artist Bernie Wrightson. Who is the illustrator of Cycle of the Werewolf? _____

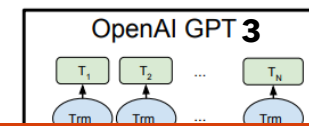
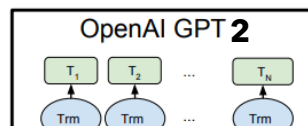
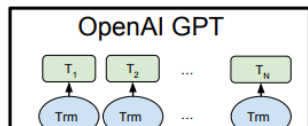
Semantic
Parsing

The table has column names... Tell me what the notes are for South Australia. What is the translation from English to SQL? _____



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Tom B. Brown [*]	Benjamin Mann [*]	Nick Ryder [*]	Melanie Subbiah [*]
Jared Kaplan [†]	Prafulla Dhariwal	Arvind Neelakantan	Pranav Shyam
Amanda Askell	Sandhini Agarwal	Ariel Herbert-Voss	Gretchen Krueger
Rewon Child	Aditya Ramesh	Daniel M. Ziegler	Jeffrey Wu
Christopher Hesse	Mark Chen	Eric Sigler	Mateusz Litwin
Benjamin Chess	Jack Clark	Christopher Berner	
Sam McCandlish	Alec Radford	Ilya Sutskever	Dario Amodei

Abstract

Recent work has demonstrated substantial gains on many NLP tasks and benchmarks by pre-training on a large corpus of text followed by fine-tuning on a specific task. While typically task-agnostic in architecture, this method still requires task-specific fine-tuning datasets of thousands or tens of thousands of examples. By contrast, humans can generally perform a new language task from only a few examples or from simple instructions - something which current NLP systems still largely struggle to do. Here we show that scaling up language models greatly improves task-agnostic, few-shot performance, sometimes even reaching competitiveness with prior state-of-the-art fine-tuning approaches. Specifically, we train GPT-3, an autoregressive language model with 175 billion parameters, 10x more than any previous non-sparse language model, and test its performance in the few-shot setting. For all tasks, GPT-3 is applied without any gradient updates or fine-tuning, with tasks and few-shot demonstrations specified purely via text interaction with the model. GPT-3 achieves strong performance on many NLP datasets, including translation, question-answering, and cloze tasks, as well as several tasks that require on-the-fly reasoning or domain adaptation, such as unscrambling words, using a novel word in a sentence, or performing 3-digit arithmetic. At the same time, we also identify some datasets where GPT-3's few-shot learning still struggles, as well as some datasets where GPT-3 faces methodological issues related to training on large web corpora. Finally, we find that GPT-3 can generate samples of news articles which human evaluators have difficulty distinguishing from articles written by humans. We discuss broader societal impacts of this finding and of GPT-3 in general.

^{*}Equal contribution
[†]Johns Hopkins University, OpenAI

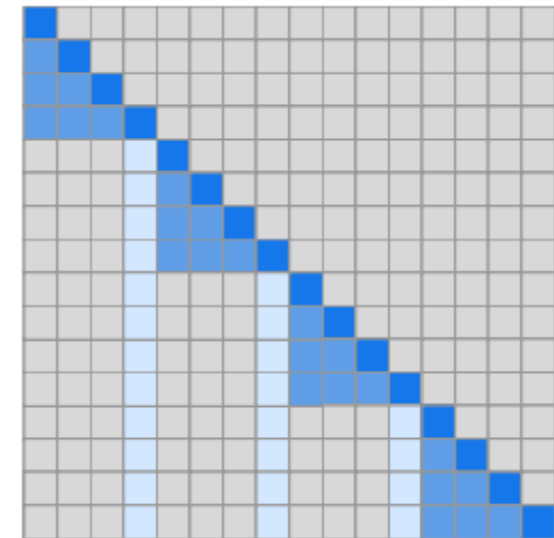
Author contributions listed at end of paper.

arXiv:2005.14165v4 [cs.CL] 22 Jul 2020



Idea: Bigger again. More data again. Stronger analysis of zero/one/few-shot.

Model Name	n_{params}	n_{layers}	d_{model}	n_{heads}	d_{head}	Batch Size	Learning Rate
GPT-3 Small	125M	12	768	12	64	0.5M	6.0×10^{-4}
GPT-3 Medium	350M	24	1024	16	64	0.5M	3.0×10^{-4}
GPT-3 Large	760M	24	1536	16	96	0.5M	2.5×10^{-4}
GPT-3 XL	1.3B	24	2048	24	128	1M	2.0×10^{-4}
GPT-3 2.7B	2.7B	32	2560	32	80	1M	1.6×10^{-4}
GPT-3 6.7B	6.7B	32	4096	32	128	2M	1.2×10^{-4}
GPT-3 13B	13.0B	40	5140	40	128	2M	1.0×10^{-4}
GPT-3 175B or “GPT-3”	175.0B	96	12288	96	128	3.2M	0.6×10^{-4}



Combined a bunch of large text corpuses into one huge training set.
Roughly 570 GB of raw text – or 400 billion tokens.

Today's Learning Objectives

Be able to answer:

- ~~What are pretrained language representation (PRL) models?~~
 - ~~How are language-model based PRL models trained?~~
 - ~~What is GPT?~~
- How to use transformers to vision tasks?
 - What is ViT?
 - What is Swin Transformer?
 - What is DETR?
 - What is MaskFormer and Segment Anything (SAM)?





Vision Transformers (ViT) as an image encoder

arXiv:2010.11929v2 [cs.CV] 3 Jun 2021

Published as a conference paper at ICLR 2021

AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE

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Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, Neil Houlsby^{*1}
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ABSTRACT

While the Transformer architecture has become the de-facto standard for natural language processing tasks, its applications to computer vision remain limited. In vision, attention is either applied in conjunction with convolutional networks, or used to replace certain components of convolutional networks while keeping their overall structure in place. We show that this reliance on CNNs is not necessary and a pure transformer applied directly to sequences of image patches can perform very well on image classification tasks. When pre-trained on large amounts of data and transferred to multiple mid-sized or small image recognition benchmarks (ImageNet, CIFAR-100, VTAB, etc.), Vision Transformer (ViT) attains excellent results compared to state-of-the-art convolutional networks while requiring substantially fewer computational resources to train.¹

1 INTRODUCTION

Self-attention-based architectures, in particular Transformers (Vaswani et al., 2017), have become the model of choice in natural language processing (NLP). The dominant approach is to pre-train on a large text corpus and then fine-tune on a smaller task-specific dataset (Devlin et al., 2019). Thanks to Transformers’ computational efficiency and scalability, it has become possible to train models of unprecedented size, with over 100B parameters (Brown et al., 2020; Lepikhin et al., 2020). With the models and datasets growing, there is still no sign of saturating performance.

In computer vision, however, convolutional architectures remain dominant (LeCun et al., 1989; Krizhevsky et al., 2012; He et al., 2016). Inspired by NLP successes, multiple works try combining CNN-like architectures with self-attention (Wang et al., 2018; Carion et al., 2020), some replacing the convolutions entirely (Ramachandran et al., 2019; Wang et al., 2020a). The latter models, while theoretically efficient, have not yet been scaled effectively on modern hardware accelerators due to the use of specialized attention patterns. Therefore, in large-scale image recognition, classic ResNet-like architectures are still state of the art (Mahajan et al., 2018; Xie et al., 2020; Kolesnikov et al., 2020).

Inspired by the Transformer scaling successes in NLP, we experiment with applying a standard Transformer directly to images, with the fewest possible modifications. To do so, we split an image into patches and provide the sequence of linear embeddings of these patches as an input to a Transformer. Image patches are treated the same way as tokens (words) in an NLP application. We train the model on image classification in supervised fashion.

When trained on mid-sized datasets such as ImageNet without strong regularization, these models yield modest accuracies of a few percentage points below ResNets of comparable size. This seemingly discouraging outcome may be expected: Transformers lack some of the inductive biases

¹Fine-tuning code and pre-trained models are available at https://github.com/google-research/vision_transformer

How do we transform an image into a set of features?

- **Just slice up the image in a 16x16 grid → 256 inputs**

How do we encode spatial locations?

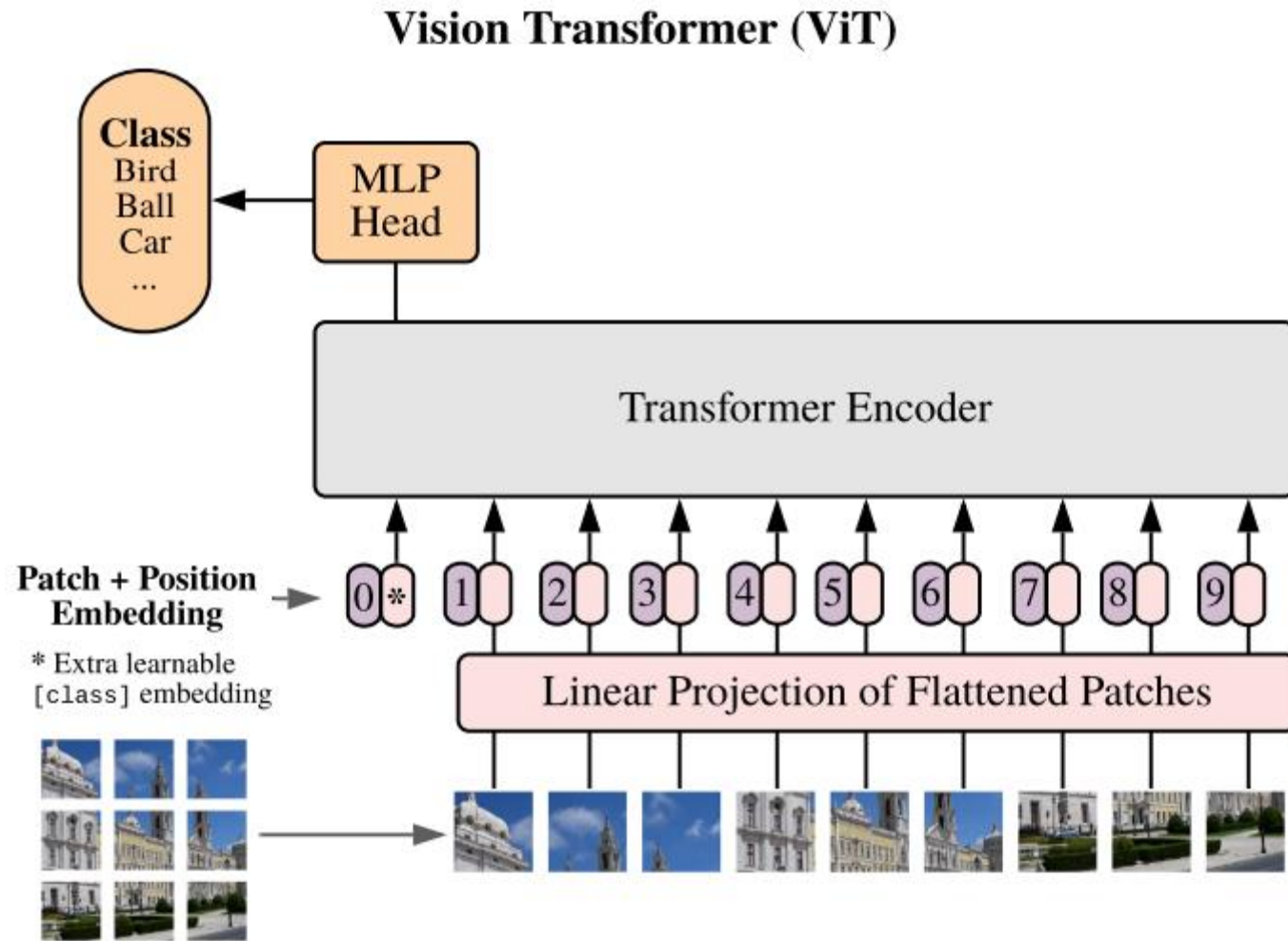
- **Learn a set of position encodings, e.g. e_{12}**

How do we get an output for something like image classification?

- **Add a learned “dummy” input. Take output representation.**



Vision Transformers (ViT) as an image encoder

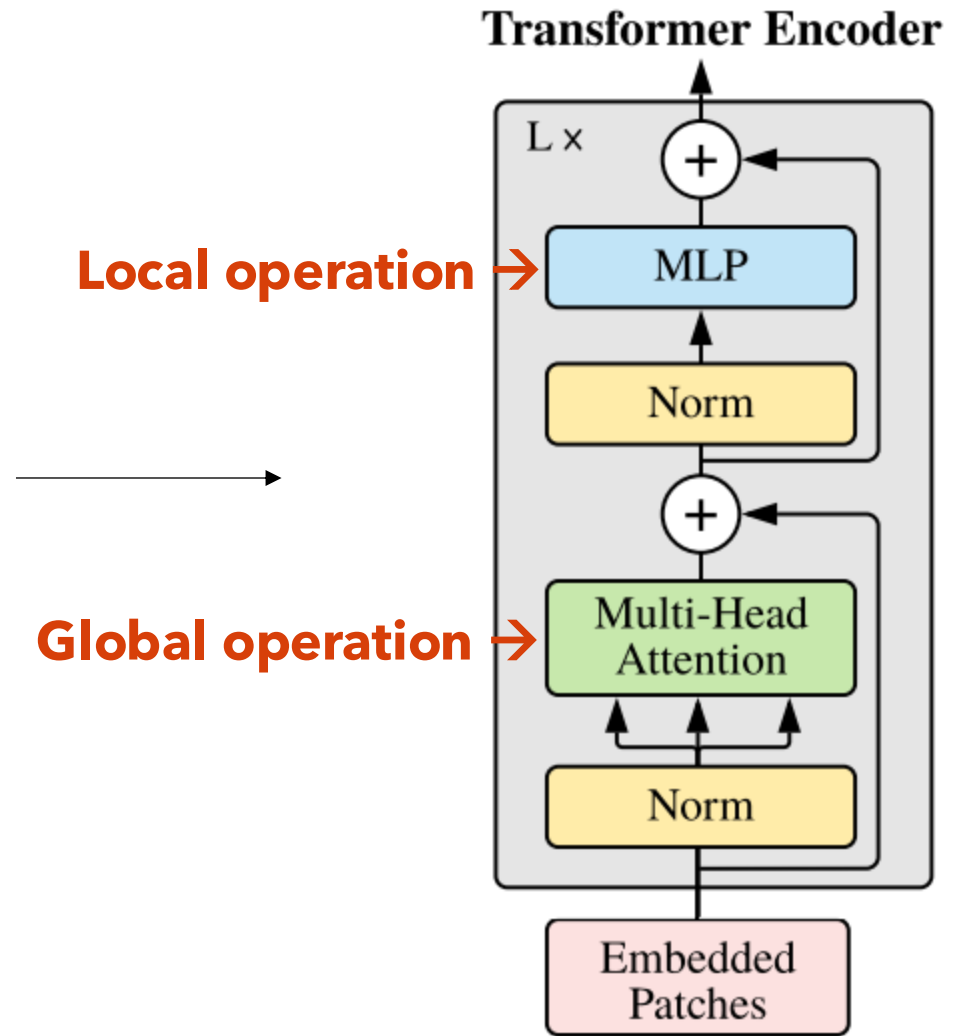
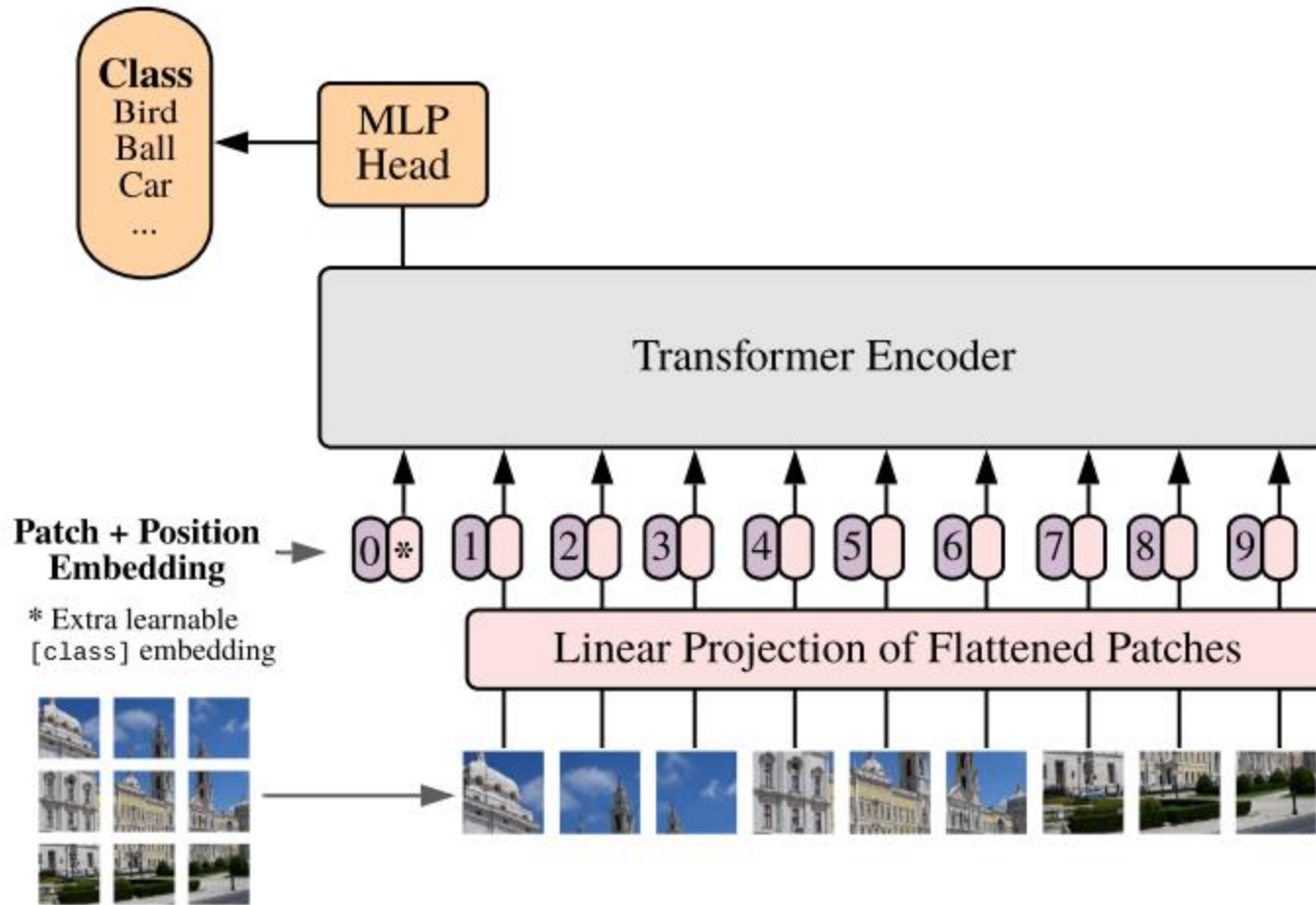


Equivalent to 16x16 conv



Vision Transformers (ViT)

Vision Transformer (ViT)



Local operation →

Global operation →

Transformer Encoder

$L \times$

+

MLP

Norm

+

Multi-Head
Attention

Norm

Embedded
Patches



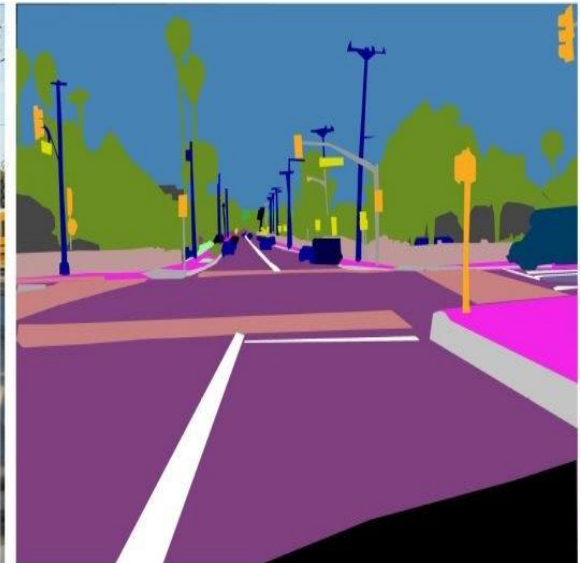
- ViT tends to be very data hungry – needs to be pretrained on HUGE datasets and then transferred.
- Hybrid ViT models where CNN process image to some spatial resolution before the transformer takes over tends to work better when huge pretraining is not possible.



Question: Is $16 \times 16 = 256$ patches enough? Probably for classification...

What about object detection?

What about segmentation? Maybe... maybe not.



Swin Transformer: Hierarchical Vision Transformer using Shifted Windows

Ze Liu^{1*} Yutong Lin^{1*} Yue Cao^{*} Han Hu^{2,3} Yixuan Wei¹
Zheng Zhang¹ Stephen Lin¹ Baining Guo¹
Microsoft Research Asia

{v-liu1, w-yutlin, yuecao, hanhu, v-yixuan, zhe2, stavelin, bainguo}@microsoft.com

Abstract

This paper presents a new vision Transformer, called Swin Transformer, that capably serves as a general-purpose backbone for computer vision. Challenges in adapting Transformer from language to vision arise from differences between the two domains, such as large variations in the scale of visual entities and the high resolution of pixels in images compared to words in text. To address these differences, we propose a hierarchical Transformer whose representation is computed with Shifted windows. The shifted windowing scheme brings greater efficiency by limiting self-attention computation to non-overlapping local windows while also allowing for cross-window connection. This hierarchical architecture has the flexibility to model at various scales and has linear computational complexity with respect to image size. These qualities of Swin Transformer make it compatible with a broad range of vision tasks, including image classification (87.3 top-1 accuracy on ImageNet-1K) and dense prediction tasks such as object detection (58.7 box AP and 51.1 mask AP on COCO test-dev) and semantic segmentation (53.5 mIoU on ADE20K val). Its performance surpasses the previous state-of-the-art by a large margin of +2.7 box AP and +2.6 mask AP on COCO, and +3.2 mIoU on ADE20K, demonstrating the potential of Transformer-based models as vision backbones. The hierarchical design and the shifted window approach also prove beneficial for all-MLP architectures. The code and models are publicly available at <https://github.com/microsoft/Swin-Transformer>.

1. Introduction

Modeling in computer vision has long been dominated by convolutional neural networks (CNNs). Beginning with AlexNet [39] and its revolutionary performance on the ImageNet image classification challenge, CNN architectures have evolved to become increasingly powerful through

^{*}Equal contribution. ¹Interns at MSRA. ²Contact person.

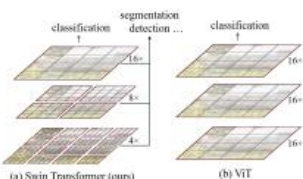


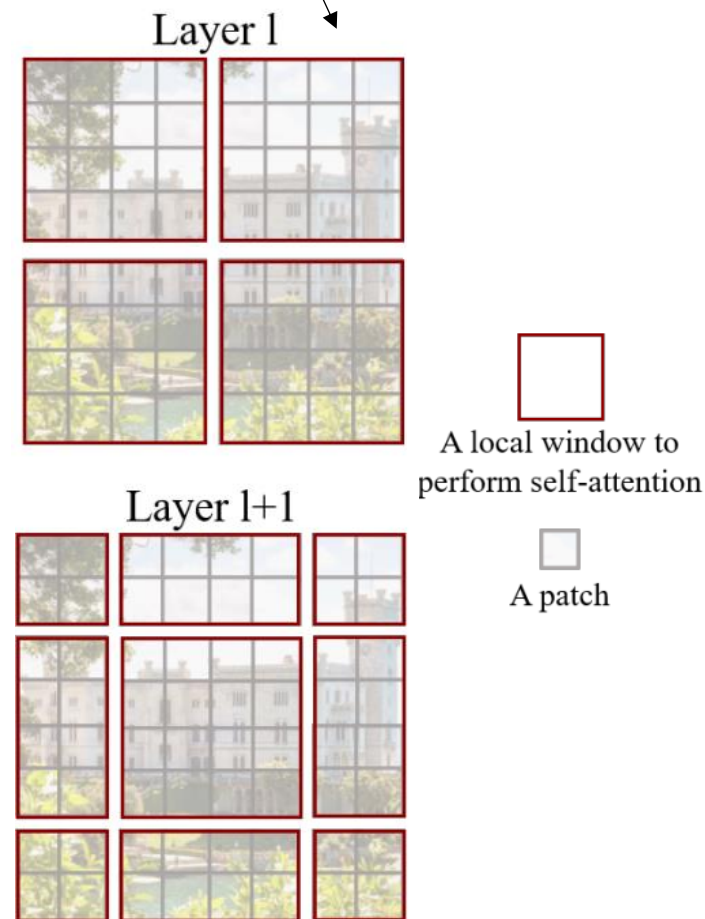
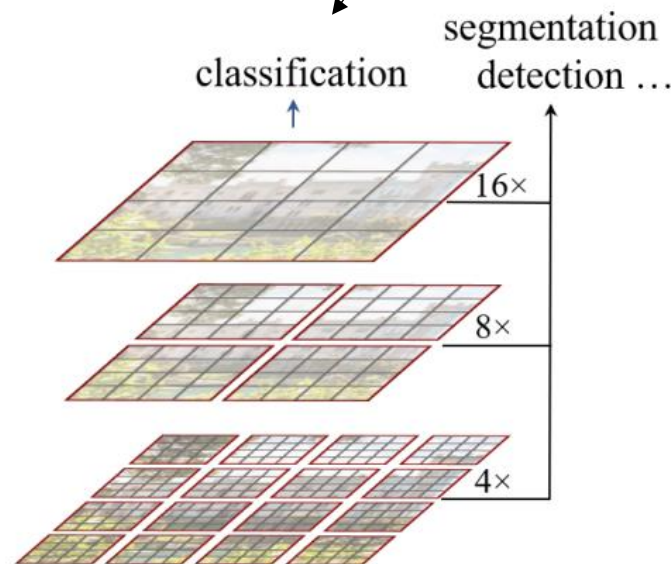
Figure 1. (a) The proposed Swin Transformer builds hierarchical feature maps by merging image patches (shown in gray) in deeper layers and has linear computation complexity to input image size due to computation of self-attention only within each local window (shown in red). It can thus serve as a general-purpose backbone for both image classification and dense recognition tasks. (b) In contrast, previous vision Transformers [39] produce feature maps of a single low resolution and have quadratic computation complexity to input image size due to computation of self-attention globally.

greater scale [30, 76], more extensive connections [34], and more sophisticated forms of convolution [70, 18, 84]. With CNNs serving as backbone networks for a variety of vision tasks, these architectural advances have led to performance improvements that have broadly lifted the entire field.

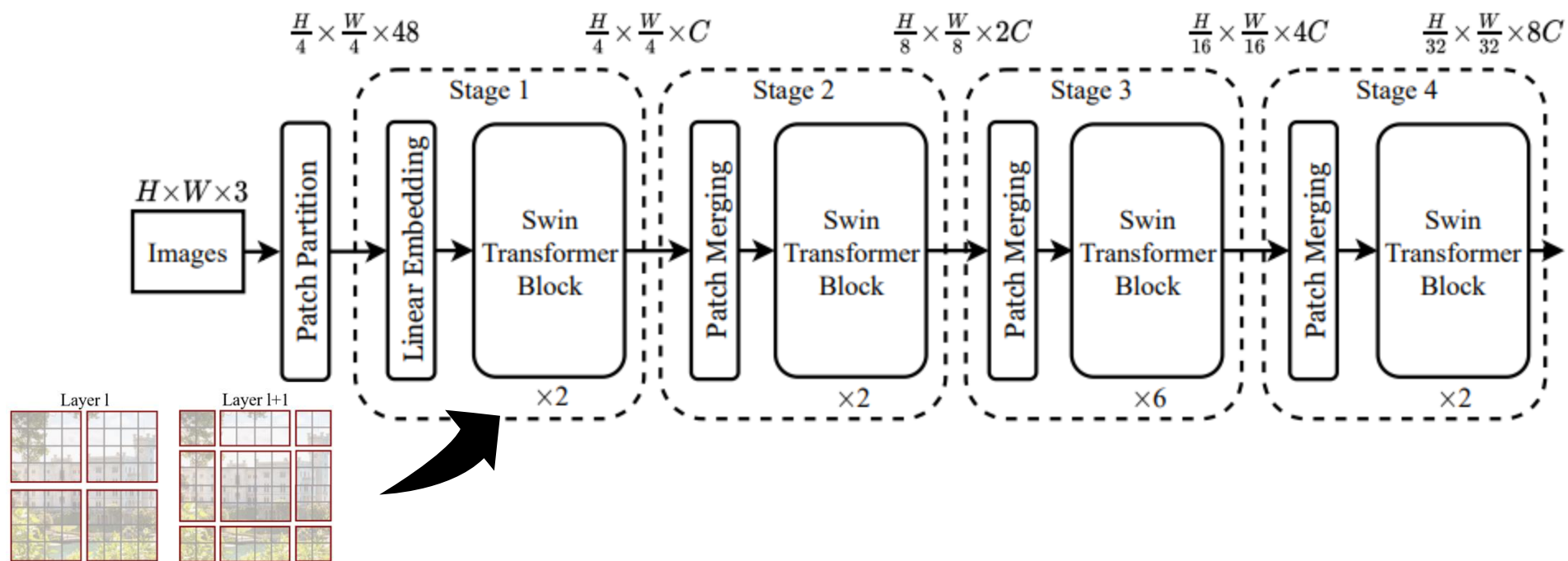
On the other hand, the evolution of network architectures in natural language processing (NLP) has taken a different path, where the prevalent architecture today is instead the Transformer [60]. Designed for sequence modeling and transduction tasks, the Transformer is notable for its use of attention to model long-range dependencies in the data. Its tremendous success in the language domain has led researchers to investigate its adaptation to computer vision, where it has recently demonstrated promising results on certain tasks, specifically image classification [20] and joint vision-language modeling [47].

In this paper, we seek to expand the applicability of Transformer such that it can serve as a general-purpose

Idea: Hierarchical model and local attention



Idea: Hierarchal model and local attention



Model basically performs interleaving of local attention operations and pooling.

Does this bias seem familiar to anyone else?



Performance of these Vision Transformers

ViTs underperform CNNs in general, but Swin Transformers are comparable/slightly better with the best CNNs

(a) Regular ImageNet-1K trained models					
method	image size	#param.	FLOPs	throughput (image / s)	ImageNet top-1 acc.
RegNetY-4G [48]	224 ²	21M	4.0G	1156.7	80.0
RegNetY-8G [48]	224 ²	39M	8.0G	591.6	81.7
RegNetY-16G [48]	224 ²	84M	16.0G	334.7	82.9
EffNet-B3 [58]	300 ²	12M	1.8G	732.1	81.6
EffNet-B4 [58]	380 ²	19M	4.2G	349.4	82.9
EffNet-B5 [58]	456 ²	30M	9.9G	169.1	83.6
EffNet-B6 [58]	528 ²	43M	19.0G	96.9	84.0
EffNet-B7 [58]	600 ²	66M	37.0G	55.1	84.3
ViT-B/16 [20]	384 ²	86M	55.4G	85.9	77.9
ViT-L/16 [20]	384 ²	307M	190.7G	27.3	76.5
DeiT-S [63]	224 ²	22M	4.6G	940.4	79.8
DeiT-B [63]	224 ²	86M	17.5G	292.3	81.8
DeiT-B [63]	384 ²	86M	55.4G	85.9	83.1
Swin-T	224 ²	29M	4.5G	755.2	81.3
Swin-S	224 ²	50M	8.7G	436.9	83.0
Swin-B	224 ²	88M	15.4G	278.1	83.5
Swin-B	384 ²	88M	47.0G	84.7	84.5

Other hybrid approaches between CNN and attention, e.g. MOAT, can perform even better



Right now? Not sure.

The ConvNeXt series of models utilize all the transformer tricks, such as large kernel sizes and layer normalization, and improves over Swin Transformers

arXiv:2201.03545v2 [cs.CV] 2 Mar 2022

A ConvNet for the 2020s

Zhuang Liu^{1,2*} Hanzi Mao¹ Chao-Yuan Wu¹ Christoph Feichtenhofer¹ Trevor Darrell² Saining Xie^{1†}

¹Facebook AI Research (FAIR) ²UC Berkeley

Code: <https://github.com/facebookresearch/ConvNeXt>

Abstract

The “Roaring 20s” of visual recognition began with the introduction of Vision Transformers (ViTs), which quickly superseded ConvNets as the state-of-the-art image classification model. A vanilla ViT, on the other hand, faces difficulties when applied to general computer vision tasks such as object detection and semantic segmentation. It is the hierarchical Transformers (e.g., Swin Transformers) that reintroduced several ConvNet priors, making Transformers practically viable as a generic vision backbone and demonstrating remarkable performance on a wide variety of vision tasks. However, the effectiveness of such hybrid approaches is still largely credited to the intrinsic superiority of Transformers, rather than the inherent inductive biases of convolutions. In this work, we reexamine the design spaces and test the limits of what a pure ConvNet can achieve. We gradually “modernize” a standard ResNet toward the design of a vision Transformer, and discover several key components that contribute to the performance difference along the way. The outcome of this exploration is a family of pure ConvNet models dubbed ConvNeXt. Constructed entirely from standard ConvNet modules, ConvNeXts compete favorably with Transformers in terms of accuracy and scalability, achieving 87.8% ImageNet top-1 accuracy and outperforming Swin Transformers on COCO detection and ADE20K segmentation, while maintaining the simplicity and efficiency of standard ConvNets.

1. Introduction

Looking back at the 2010s, the decade was marked by the monumental progress and impact of deep learning. The primary driver was the renaissance of neural networks, particularly convolutional neural networks (ConvNets). Through the decade, the field of visual recognition successfully shifted from engineering features to designing (ConvNet) architectures. Although the invention of back-propagation-trained ConvNets dates all the way back to the 1980s [42], it was not until late 2012 that we saw its true potential for

*Work done during an internship at Facebook AI Research.

†Corresponding author.

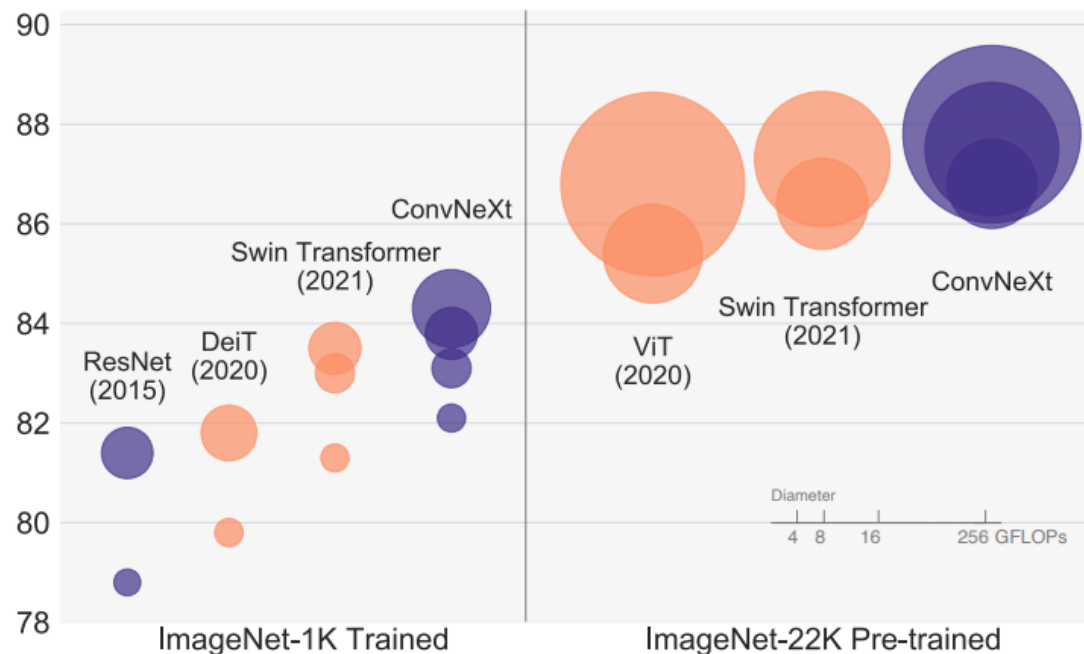


Figure 1. ImageNet-1K classification results for ConvNets and vision Transformers. Each bubble’s area is proportional to FLOPs of a variant in a model family. ImageNet-1K/22K models here take 224²/384² images respectively. ResNet and ViT results were obtained with improved training procedures over the original papers. We demonstrate that a standard ConvNet model can achieve the same level of scalability as hierarchical vision Transformers while being much simpler in design.

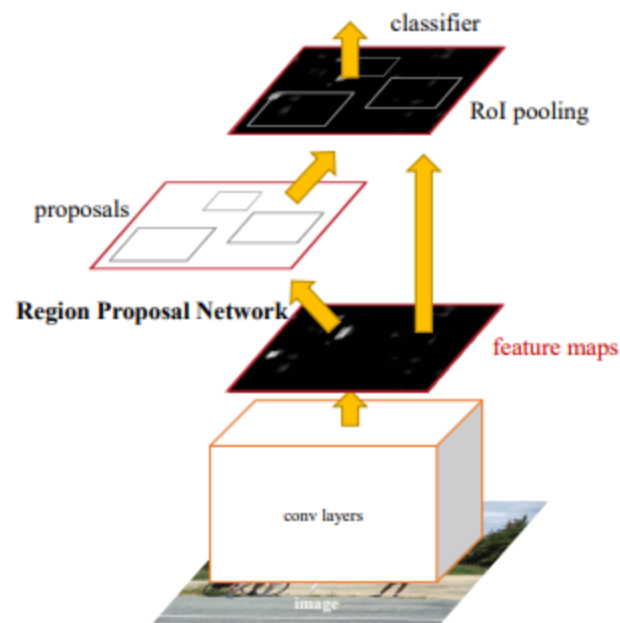
visual feature learning. The introduction of AlexNet [40] precipitated the “ImageNet moment” [59], ushering in a new era of computer vision. The field has since evolved at a rapid speed. Representative ConvNets like VGGNet [64], Inception [68], ResNe(X)t [28, 87], DenseNet [36], MobileNet [34], EfficientNet [71] and RegNet [54] focused on different aspects of accuracy, efficiency and scalability, and popularized many useful design principles.

The full dominance of ConvNets in computer vision was not a coincidence: in many application scenarios, a “sliding window” strategy is intrinsic to visual processing, particularly when working with high-resolution images. ConvNets have several built-in inductive biases that make them well-suited to a wide variety of computer vision applications. The most important one is translation equivariance, which is a desirable property for tasks like object detection. ConvNets are also inherently efficient due to the fact that when used in a sliding-window manner, the computations are shared [62]. For many decades, this has been the default use of ConvNets, generally on limited object categories such as digits [43], faces [58, 76] and pedestrians [19, 63]. Entering the 2010s,

ImageNet-1K Acc.



Swin Transformer can be a drop-in replacement as a vision **backbone**.



Recall Faster RCNN. Just swap out “conv layers” (from a ResNet) to Swin Transformer.

Today's Learning Objectives

Be able to answer:

- ~~What are pretrained language representation (PRL) models?~~
 - ~~How are language-model based PRL models trained?~~
 - ~~What is GPT?~~
- ~~How to use transformers to vision tasks?~~
 - ~~What is ViT?~~
 - ~~What is Swin Transformer?~~
 - What is DETR?
 - What is MaskFormer and Segment Anything (SAM)?





End-to-End Object Detection with Transformers

Nicolas Carion*, Francisco Massa*, Gabriel Synnaeve, Nicolas Usunier, Alexander Kirillov, and Sergey Zagoruyko

Facebook AI

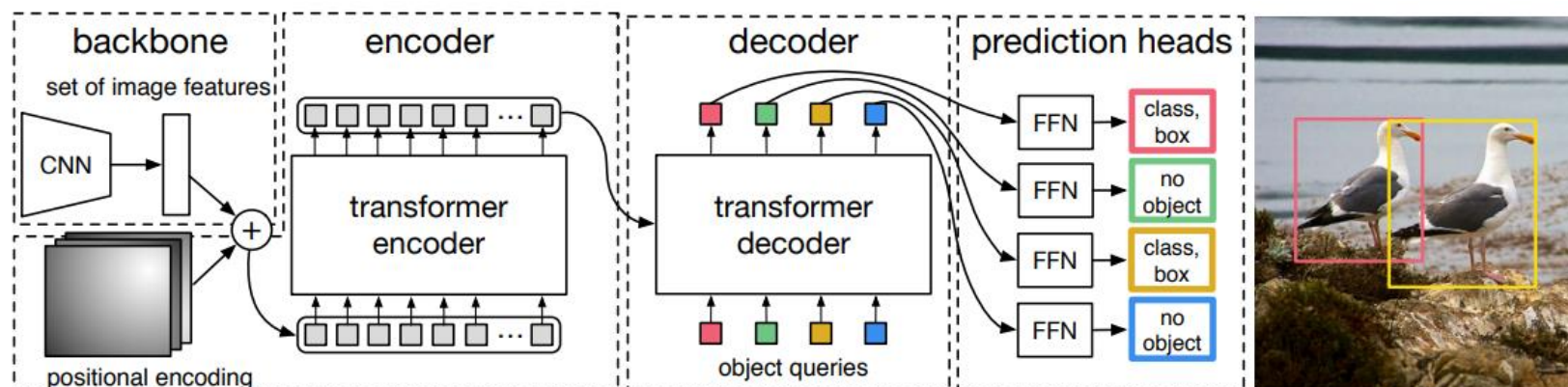
Abstract. We present a new method that views object detection as a direct set prediction problem. Our approach streamlines the detection pipeline, effectively removing the need for many hand-designed components like a non-maximum suppression procedure or anchor generation that explicitly encode our prior knowledge about the task. The main ingredients of the new framework, called DETR (DEtection TRansformer or DeTR), are a set-based global loss that forces unique predictions via bipartite matching, and a transformer encoder-decoder architecture. Given a fixed small set of learned object queries, DETR reasons about the relations of the objects and the global image context to directly output the final set of predictions in parallel. The new model is conceptually simple and does not require a specialized library, unlike many other modern detectors. DETR demonstrates accuracy and run-time performance on par with the well-established and highly-optimized Faster R-CNN baseline on the challenging COCO object detection dataset. Moreover, DETR can be easily generalized to produce panoptic segmentation in a unified manner. We show that it significantly outperforms competitive baselines. Training code and pretrained models are available at <https://github.com/facebookresearch/detr>.

1 Introduction

The goal of object detection is to predict a set of bounding boxes and category labels for each object of interest. Modern detectors address this set prediction task in an indirect way, by defining surrogate regression and classification problems on a large set of proposals [37, 5], anchors [23], or window centers [53, 46]. Their performances are significantly influenced by postprocessing steps to collapse near-duplicate predictions, by the design of the anchor sets and by the heuristics that assign target boxes to anchors [52]. To simplify these pipelines, we propose a direct set prediction approach to bypass the surrogate tasks. This end-to-end philosophy has led to significant advances in complex structured prediction tasks such as machine translation or speech recognition, but not yet in object detection: previous attempts [43, 16, 4, 39] either add other forms of prior knowledge, or have not proven to be competitive with strong baselines on challenging benchmarks. This paper aims to bridge this gap.

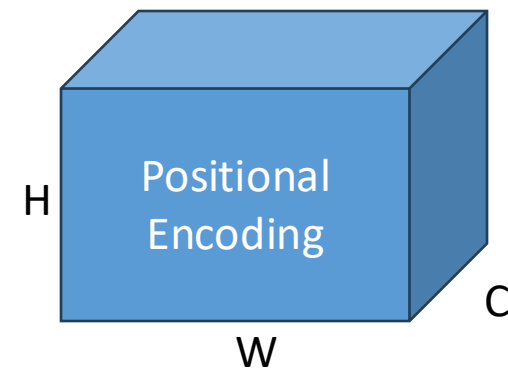
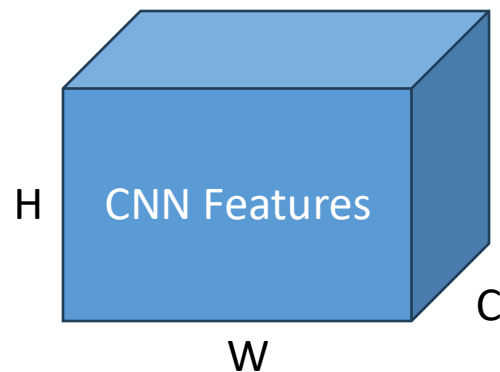
* Equal contribution

Idea: Instead of using anchor boxes, maybe we can learn “ideas” about box proposals as tokens and use an attention model?

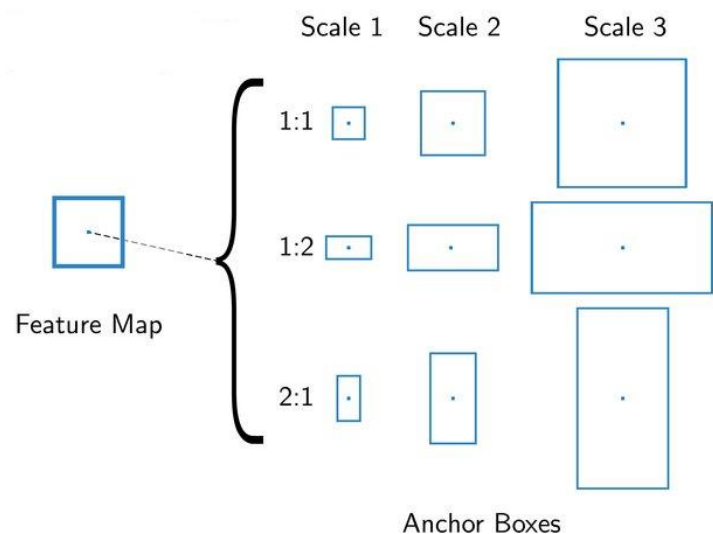




Contrast between Anchor Boxes and Learned Proposal Tokens



Anchor Boxes



Reflects prior information about
"where to look for boxes"

Transformer Decoder

Learned "Object Query" Tokens



Learns about
"where to look for object boxes"

Fully-connected to the entire image
to find activations

Can replicate what anchor boxes can do
by attend to positional information

Can also learn to look for certain features



What DETR outputs

Learned "Object" Query Tokens



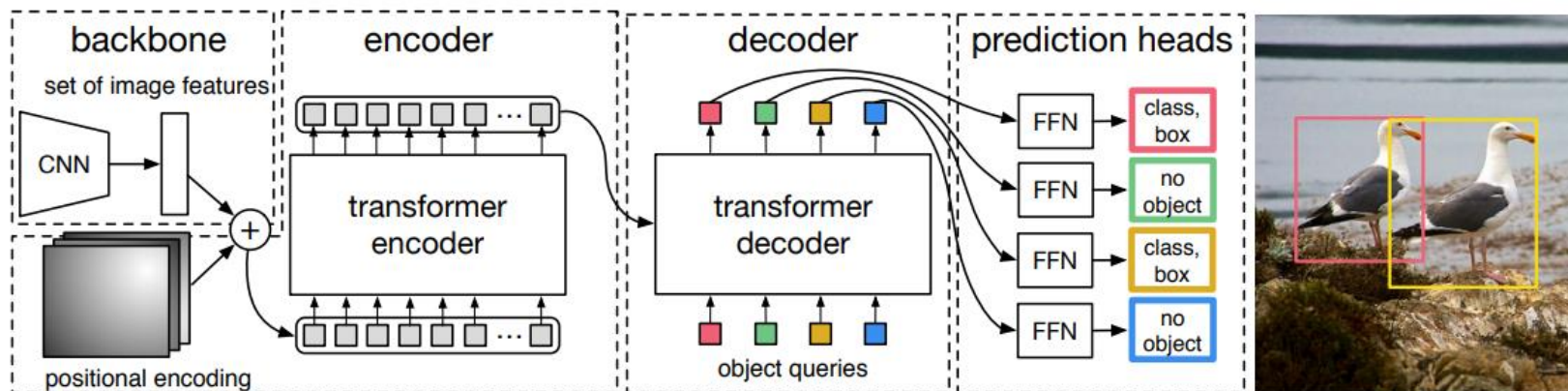
One box/category from each learned object query

What if two boxes match the same query?

-- Too bad you are unlucky. Or, we can learn more queries.

What if no box match the query because they are in a slightly different location?

-- Too bad you are unlucky. Or, we can learn more queries.



How many queries? 100 - 900



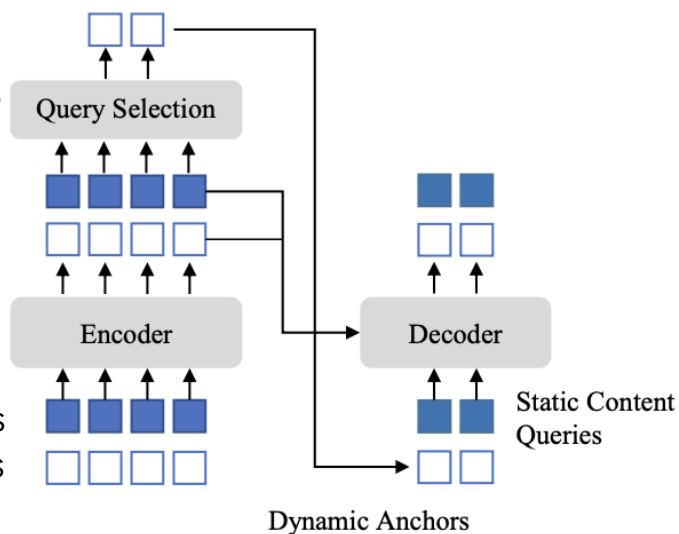
DINO: Combining Anchor Box Idea with Learned Content Queries

Basic idea: We can still search the image using the anchor box idea, so we only need to learn the “content features” and a position offset

An RPN layer outputs a score for each location, top-k is selected with predicted position offsets

These are the tokens that the decoder “attends to”

Backbone Features
Positional Embeddings



(c) Mixed Query Selection

DINO: DETR with Improved DeNoising Anchor Boxes for End-to-End Object Detection

Hao Zhang^{1,3*†}, Feng Li^{1,3*†}, Shilong Liu^{2,3*†}, Lei Zhang^{3‡},
Hang Su², Jun Zhu², Lionel M. Ni^{1,4}, Heung-Yeung Shum^{1,3}

¹The Hong Kong University of Science and Technology.

²Dept. of CST., BNRist Center, Institute for AI, Tsinghua University.

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{ni, hshum}@ust.hk

{leizhang}@idea.edu.cn

Those are learned during the training phase

Method	Params	Backbone Pre-training Dataset	Detection Pre-training Dataset	Use Mask	End-to-end	val2017 (AP)		test-dev (AP)	
						w/o TTA	w/ TTA	w/o TTA	w/ TTA
SwinL [23]	284M	IN-22K-14M	O365	✓		57.1	58.0	57.7	58.7
DyHead [7]	≥ 284M	IN-22K-14M	Unknown*			—	58.4	—	60.6
Soft Teacher+SwinL [38]	284M	IN-22K-14M	O365	✓		60.1	60.7	—	61.3
GLIP [18]	≥ 284M	IN-22K-14M	FourODs [18], GoldG+ [18,15]			—	60.8	—	61.5
Florence-CoSwin-H[40]	≥ 637M	FLD-900M [40]	FLD-9M [40]			—	62.0	—	62.4
SwinV2-G [22]	3.0B	IN-22K-ext-70M [22]	O365	✓		61.9	62.5	—	63.1
DINO-SwinL(Ours)	218M	IN-22K-14M	O365		✓	63.1	63.2	63.2	63.3

Table 3. Comparison of the best detection models on MS-COCO. Similar to DETR [3],



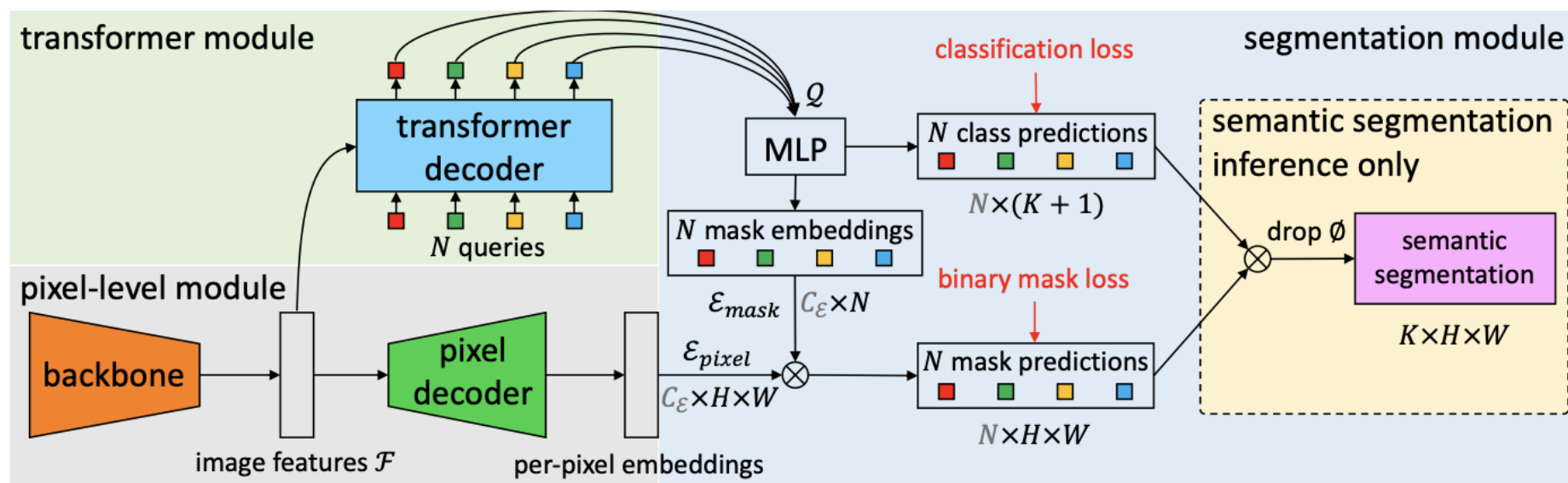
Per-Pixel Classification is Not All You Need for Semantic Segmentation

- Similar idea: 1) Use transformer to obtain multiple proposals
- 2) Attend to all pixels for segmentation results

Bowen Cheng^{1,2*} Alexander G. Schwing² Alexander Kirillov¹

¹Facebook AI Research (FAIR)

²University of Illinois at Urbana-Champaign (UIUC)





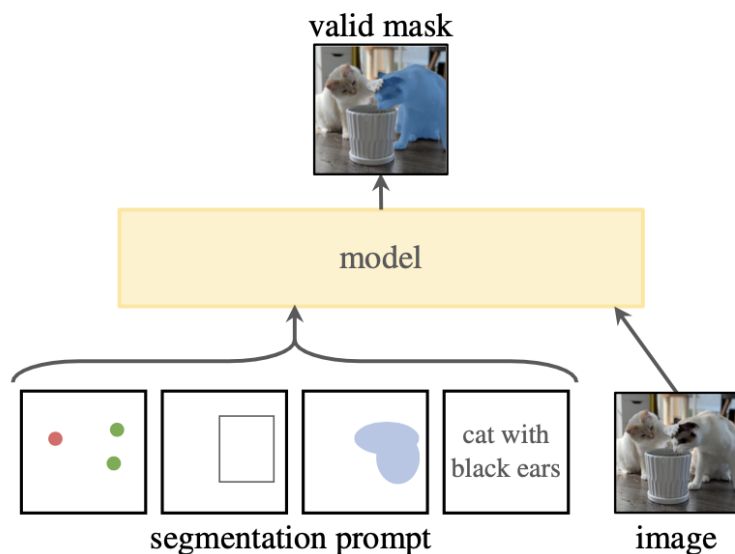
Segment Anything (SAM)

Segment Anything

Alexander Kirillov^{1,2,4} Eric Mintun² Nikhila Ravi^{1,2} Hanzi Mao² Chloe Rolland³ Laura Gustafson³
Tete Xiao³ Spencer Whitehead Alexander C. Berg Wan-Yen Lo Piotr Dollár⁴ Ross Girshick⁴
¹project lead ²joint first author ³equal contribution ⁴directional lead

Meta AI Research, FAIR

Idea: Generate tokens from different types of prompts



(a) **Task:** promptable segmentation

Model: Similar to MaskFormer

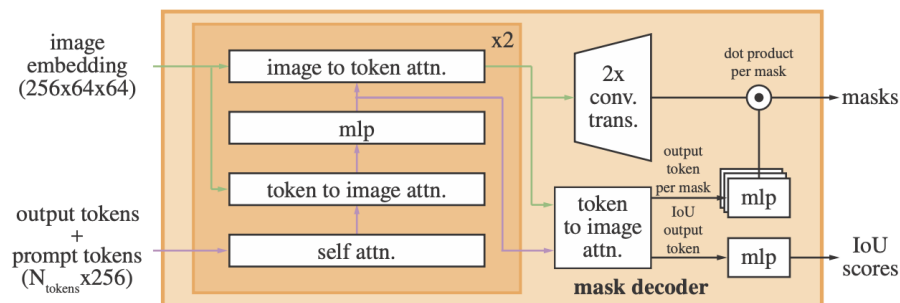


Figure 14: Details of the lightweight mask decoder. A two-layer decoder updates both the image embedding and prompt tokens via cross-attention. Then the image embedding is upsampled, from which the updated output tokens are used to dynamically predict masks. (Not illustrated for figure clarity: At every attention layer, positional encodings are added to the image embedding, and the entire original prompt token (including position encoding) is re-added to the token queries and keys.)



<https://segment-anything.com/>

Today's Learning Objectives

Be able to answer:

- ~~What are pretrained language representation (PRL) models?~~
 - ~~How are language-model based PRL models trained?~~
 - ~~What is GPT?~~
 - ~~How can we use these models once they are trained?~~
- ~~How to use transformers to vision tasks?~~
 - ~~What is ViT?~~
 - ~~What is Swin Transformer?~~
 - ~~What is DETR?~~
 - ~~What is MaskFormer and Segment Anything (SAM)?~~

